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命题转化

1.1 优势腾挪

“腾”思想，正是数学家们面对复杂问题时的核心策略之一。当一个问题原始设定带有令人不悦的限制、非标准的环境或复杂的局部结构时，高手们想的不是硬着头皮在原地解决，而是：

“我能否构建一个新的世界，在那个世界里，这些障碍都烟消云散，变成了新世界的基本法则”

Theorem 1.1.1

Suppose φ is an **Borel outer measure** on a metric space X . Then for each Borel set $B \subseteq X$ with $\varphi(B) < \infty$ and each $\varepsilon > 0$, there exists a closed set $F \subseteq B$ such that

$$\varphi(B \setminus F) < \varepsilon$$

Furthermore, suppose

$$B \subseteq \bigcup_{i=1}^{\infty} V_i$$

where each V_i is an open set with $\varphi(V_i) < \infty$. Then for each $\varepsilon > 0$, there is an open set $W \supseteq B$ such that

$$\varphi(W \setminus B) < \varepsilon$$

Note.

题目对于 B 的条件给予了“额外”(脱离 σ -代数体系) 的增强条件 $\varphi(B) < \infty$. 通过定义新的测度 $\mu(A) := \varphi(A \cap B)$. 将增强条件“腾挪”到变成测度的增强条件 $\mu(X) < \infty$. 以增加命题的整体性，使得证明不拘泥于某个特定集合的 $\varphi(B) < \infty$.

Proof. Select a Borel set B with $\varphi(B) < \infty$, and define

$$\mu(A) := \varphi(A \cap B)$$

It is easy to verify that μ is an outer measure on X , and also all Borel set is

μ -measurable. Furthermore, $\mu(X) < \infty$. Once we show that the first part is true for μ , then for each $\varepsilon > 0$

$$\varphi(B \setminus F) = \mu(B \setminus F) < \varepsilon$$

is true for some closed set F and the specific Borel set B we have just selected. Since B is arbitray, we may assume that $\varphi(X) < \infty$ in the following process. Let

$$\mathcal{D} := \{A : \forall \varepsilon > 0, \exists \text{ closed set } F \subseteq A \text{ s.t. } \varphi(A \setminus F) < \varepsilon\}$$

Only need to show that $\mathcal{D}$ is a σ -algebra containing all closed sets. That is to say from Theorem ?? that $\mathcal{D}$ contains all closed and open sets and is closed under countable union and countable intersection. Take any open set $U \subseteq X$. Let

$$F_i := \left\{x : d(x, U^c) \geq \frac{1}{i}\right\}$$

be a closed set for each i . Then

$$F_1 \subseteq F_2 \subseteq \dots, \quad U = \bigcup_{i=1}^{\infty} F_i$$

Thus

$$\lim_{k \rightarrow \infty} \varphi\left(U \setminus \bigcup_{i=1}^k F_i\right) = \varphi\left(U \setminus \bigcup_{i=1}^{\infty} F_i\right) = \varphi(\emptyset) = 0$$

For each $\varepsilon > 0$, there exists k such that

$$\varphi\left(U \setminus \bigcup_{i=1}^k F_i\right) < \varepsilon$$

, where $\bigcup_{i=1}^k F_i$ is a closed set.

Now, we take a sequence of membser $A_1, A_2, \dots$ in $\mathcal{D}$. For each i , there is a closed set $C_i \subseteq A_i$, such that

$$\varphi(A_i \setminus C_i) < \frac{\varepsilon}{2^{i+1}}$$

Note that

$$\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{\infty} C_i \subseteq \bigcup_{i=1}^{\infty} (A_i \setminus C_i), \quad \bigcap_{i=1}^{\infty} A_i \setminus \bigcap_{i=1}^{\infty} C_i \subseteq \bigcup_{i=1}^{\infty} (A_i \setminus C_i)$$

We have

$$\begin{aligned} \varphi\left(\bigcap_{i=1}^{\infty} A_i \setminus \bigcap_{i=1}^{\infty} C_i\right) &\leq \sum_{i=1}^{\infty} \varphi(A_i \setminus C_i) < \varepsilon \\ \lim_{k \rightarrow \infty} \varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^k C_i\right) &= \varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{\infty} C_i\right) \leq \sum_{i=1}^{\infty} \varphi(A_i \setminus C_i) < \frac{\varepsilon}{2} \end{aligned}$$

Then there exists k such that

$$\varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^k C_i\right) < \varepsilon$$

Thus $\bigcap_{i=1}^{\infty} A_i, \bigcup_{i=1}^{\infty} A_i \in \mathcal{D}$. □